

How Well Do LLMs Predict Human Behavior? A Measure of their Pretrained Knowledge

Wayne Gao
(Penn)

Sukjin Han
(Bristol)

Annie Liang
(Northwestern)

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Introduction

A striking feature of LLMs:

- mimic human behaviors across a wide range of domains
- because of extensive **pretraining** on human-generated text
- e.g., LLMs reproduce human behavioral biases (anchoring, framing, loss aversion) (Jones & Steinhardt 22, Horton 23)

Thus increasingly used to **predict** human behaviors, e.g.,

- predicting outcomes of economic experiments (Aher et al 23, Hewitt et al 24, Shapira et al 25, Park et al 24)
- forecasting macroeconomic variables (Bybee 23, Faria-e-Castro and Leibovici 24, Carriero et al 25)

Example

- Fix a lottery: $\bar{z}$ with probability p , $\underline{z}$ with probability $1 - p$
- What is an agent's **certainty equivalent** for this lottery?
 - i.e., the sure dollar amount that is considered equivalent to the random outcome of the lottery



ECONOMIST

The agent is an **expected utility maximizer** with utility function $v_\theta(z) = z^\theta$ where θ governs the degree of risk aversion.

In this model, the certainty equivalent of lottery $(\bar{z}, p; \underline{z}, 1 - p)$ is

$$v_\theta^{-1} \left(p v_\theta(\bar{z}) + (1 - p) v_\theta(\underline{z}) \right)$$

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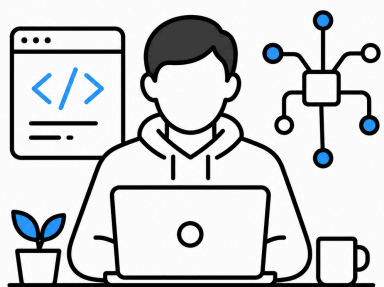
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Use data of lotteries and certainty equivalents to estimate θ , and predict the implied certainty equivalent.

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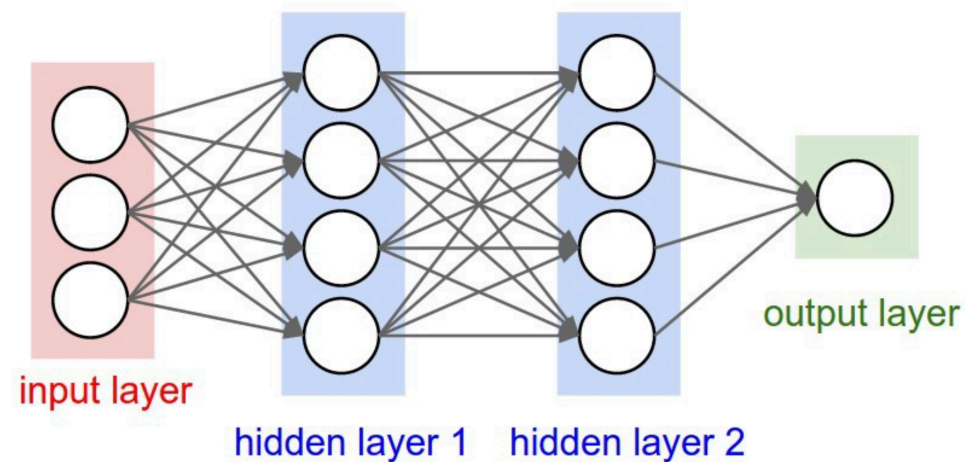
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COMPUTER
SCIENTIST

Use data of lotteries and certainty equivalents to train a machine learning algorithm (e.g., a neural net).

lottery
 $(\bar{z}, p; \underline{z}, 1 - p)$



certainty
equivalent y



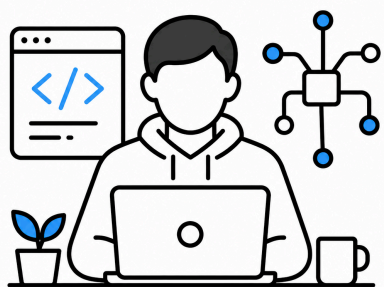
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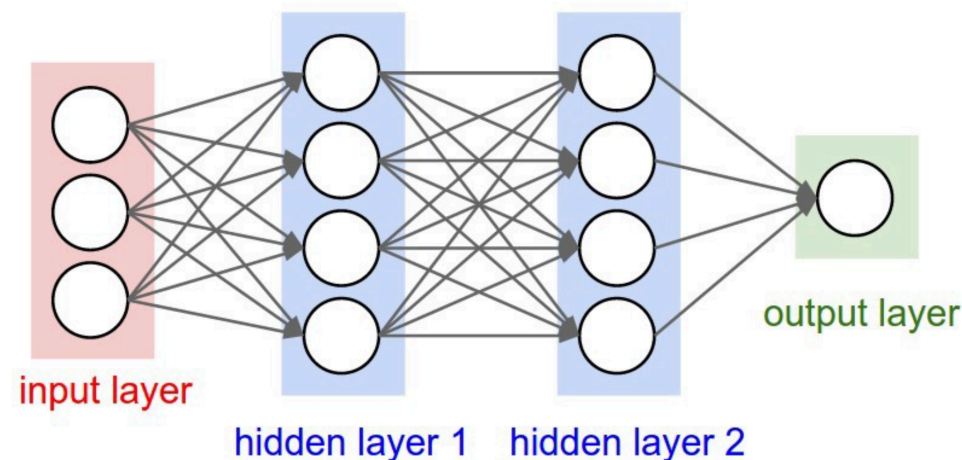
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User of LLMs

A subject is offered a lottery where they receive 10 dollars with probability 1/2 and 0 dollars with probability 1/2. What is the dollar amount x such that this subject would be indifferent between receiving the random outcome of the lottery, versus x dollars for sure? Your answer must be a single number.



Thought for a few seconds

5

Example

NO DOMAIN-SPECIFIC DATA REQUIRED!

...but are these predictions any good?



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Proposed Measure

- We need a way to evaluate how useful the LLM is
- One option is simply “predictive accuracy”
- Two issues with this:
 - Predictive accuracy can be hard to interpret (is 2.4 RMSE small?)
 - The LLM exists in the context of alternative approaches to prediction, so its **relative** value is often what is relevant
- Key tradeoff: with enough domain-specific data, flexible algorithms will outperform the LLM, but that data is costly
- Our proposed measure: the LLM’s **equivalent sample size**, defined as how much data is needed for the algorithm to beat the LLM
- Can be interpreted as a proxy for “cost saved”

This Talk

1. Framework and proposed measure
2. Statistical inference procedure
 - compare the LLM's prediction error to that of flexible ML algorithms/economic models trained on samples of increasing size
3. Applications
 - experimental behaviors: risk preferences, initial play in games
 - survey outcomes in PSID: wages, home-ownership, smoking, etc.

Framework

Prediction Problems

Generic **prediction problem**:

- covariates $X \in \mathcal{X}$
- outcome of interest $Y \in \mathcal{Y}$ (could be a distribution)

A **prediction rule** is a mapping $f: \mathcal{X} \rightarrow \mathcal{Y}$

Prediction Problems

Measure prediction accuracy via a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$

- e.g. squared error $\ell(y, \hat{y}) = (y - \hat{y})^2$ (regression)
- e.g., misclassification $\ell(y, \hat{y}) = 1\{\hat{y} \neq y\}$ (classification)

The **risk** (expected loss) of prediction rule f is

$$e(f) := \mathbb{E}_P[\ell(f(X), Y)]$$

where P is the distribution of (X, Y)

LLM Prediction Rule

Fix an LLM and prompting protocol:

- Describe the prediction task
- Give the LLM X in natural language and ask it to predict Y

Denote the resulting prediction rule by f_{LLM} and its risk by $e_{LLM} \equiv e(f_{LLM})$

Domain-Specific Learning

An **algorithm** (or **economic model**) maps data to a prespecified set of prediction rules:

$$a : \mathcal{D} \rightarrow \mathcal{F}$$

where $\mathcal{D} \equiv \cup_{N \geq 1} (\mathcal{X} \times \mathcal{Y})^N$ is the set of all finite datasets.

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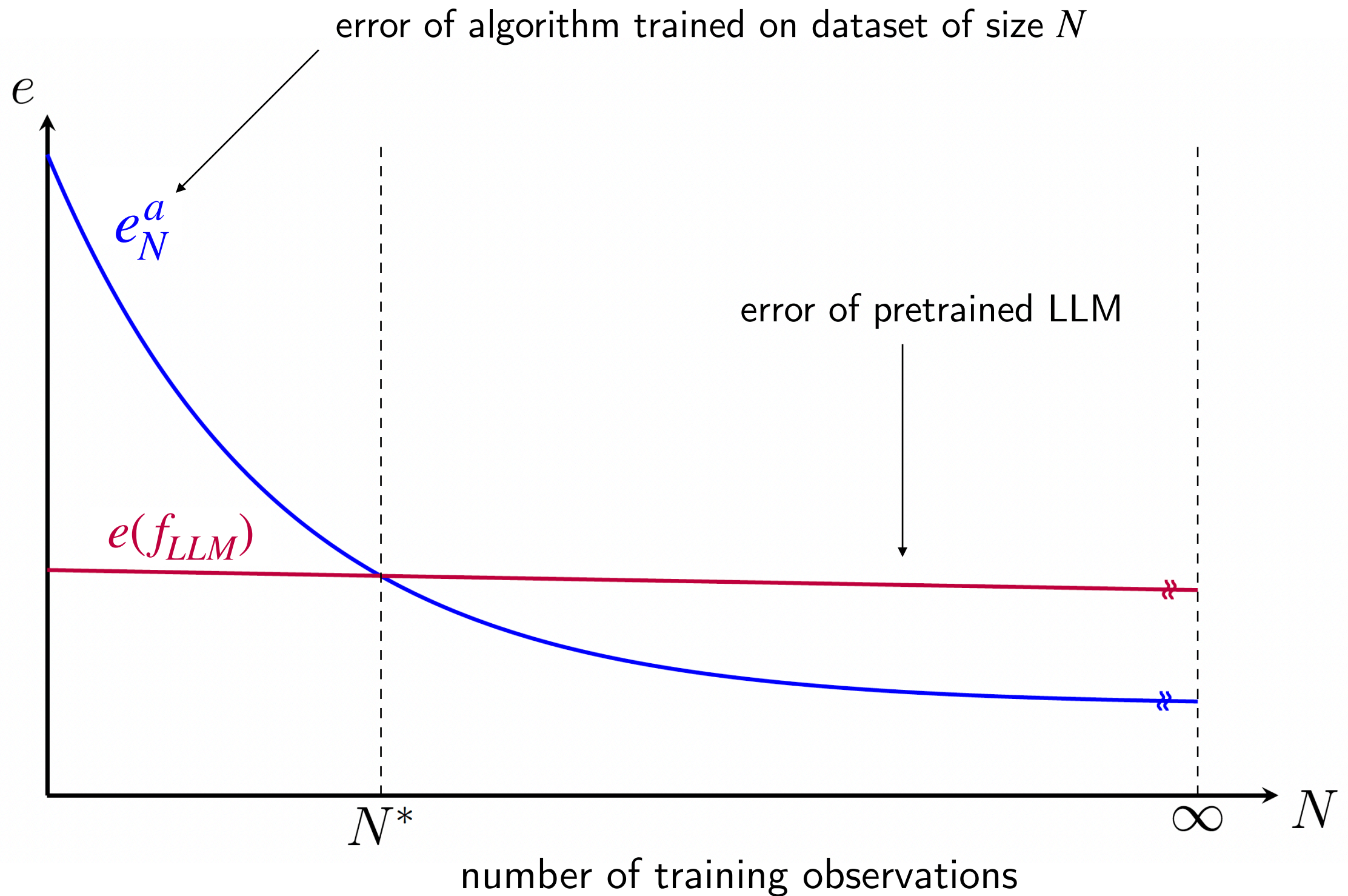
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Given sample $D_N \equiv \{(x_i, y_i)\}_{i=1}^N$ define $f_N \equiv a(D_N)$ to be the (random) prediction rule we get from applying a to a (random) dataset of size N .

Define $e_N^a \equiv \mathbb{E}[e(f_N)] = \mathbb{E}_P[\ell(f_N(X), Y)]$ to be the expected risk of f_N .

Domain-Specific Error Curve



α -Equivalent Sample Size (ESS)

Definition: The LLM's α -equivalent sample size (ESS) is

$$N^* = \min\{N : e_N^\alpha \leq e_{LLM}\}$$

- smallest training size such that the algorithm/model outperforms the pretrained LLM
- interpret as the minimum amount of domain-specific data the analyst would need to beat the LLM
- large $N^* \Rightarrow$ the LLM provides substantial predictive value in this domain; small $N^* \Rightarrow$ it does not

α -Equivalent Sample Size (ESS)

ESS depends on:

- The LLM and prompting protocol
- The problem domain
- The comparator algorithm α

Varying these and computing ESS for different choices can be instructive, as we will see in the applications

Estimation and Inference for N^*

Estimating N^*

In the paper, we propose a method for estimating and doing inference on N^* given a sample of (x, y) observations (skip details today)

- Estimate e_{LLM} by computing empirical risk on that sample
- Estimate e_N^a using block-out cross-validation:
 - Split the sample into blocks of size N , train on one block and test on the remaining $N - 1$, average across blocks
 - Characterize asymptotic distribution of this blockout cross-validated estimator

Plugin estimator for N^* : first N at which our estimate of e_N^a beats our estimate of e_{LLM} . Show that this is **consistent**.

Inference on N^*

- Formulate a sequence of hypotheses tests:

$$H_{0,N} : e_N^a \leq e_{LLM} \quad N = 1, 2, \dots$$

i.e., the algorithm trained on N observations beats the LLM

- We provide a valid statistical test for each null $H_{0,n}$ at level α
- Then propose a sequential procedure:
 - Test $H_{0,1}$ at significance level α , if reject then move onto $H_{0,2}$, etc...
 - At first N at which we cannot reject, stop and set $\widehat{N}_\alpha^* = N$
- Finally argue that validity of the individual tests implies that $\left[\widehat{N}_\alpha^*, \infty \right)$ is a **valid** one-sided confidence interval for N^* at level α

Application 1:

Predicting Risk Preferences

Setting

Prediction problem: Given a country and a lottery, predict the average certainty equivalent reported by subjects in that country

- e.g., “what are people willing to pay (on average) for the outcome of a lottery that is equally likely to yield \$100 or \$0?”

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Data: average certainty equivalents for 28 binary lotteries from subjects in 30 countries (l'Haridon & Vieider 19)

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- Y is the average certainty equivalent

Our question: How useful are LLMs for predicting certainty equivalents, and how does this vary across countries?

- We treat each of the 30 countries as a separate dataset
- Evaluate the LLM's N^* for each country's prediction problem

Cross-Country Data and LLM Predictions

LLM prediction: use GPT-5.4 to predict an average certainty equivalent for each lottery

In a cross-country experiment, participants in {country} stated the guaranteed amount that would make them indifferent between receiving that amount for sure and playing a lottery.

For the lottery below, simulate certainty equivalents for 200 synthetic subjects drawn from the participant pool in {country}. Treat the draws as different plausible respondents, not repeated copies of the same person. Allow for realistic heterogeneity across subjects in how they respond, but do not force the draws to be evenly spaced or artificially diverse. At the same time, do not collapse the distribution to nearly identical values.

Lottery:

- Outcome 1: {z_1} with probability {p_1}
- Outcome 2: {z_2} with probability {1-p_1}

Return exactly 200 numeric certainty equivalents. Each draw should be a plausible individual response for one subject.

LLM prediction:
the average of the
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Cross-Country Data and LLM Predictions

LLM prediction: use GPT-5.4 to predict an average certainty equivalent for each lottery

Domain-specific comparators: we consider a random forest algorithm and two popular economic models

- Expected Utility with utility function $u(z) = z^\alpha$ for gains, $u(z) = -(-z)^\beta$ for losses
- Cumulative Prospect Theory: agents maximize expected utility but distort probabilities

Cross-Country Data and LLM Predictions

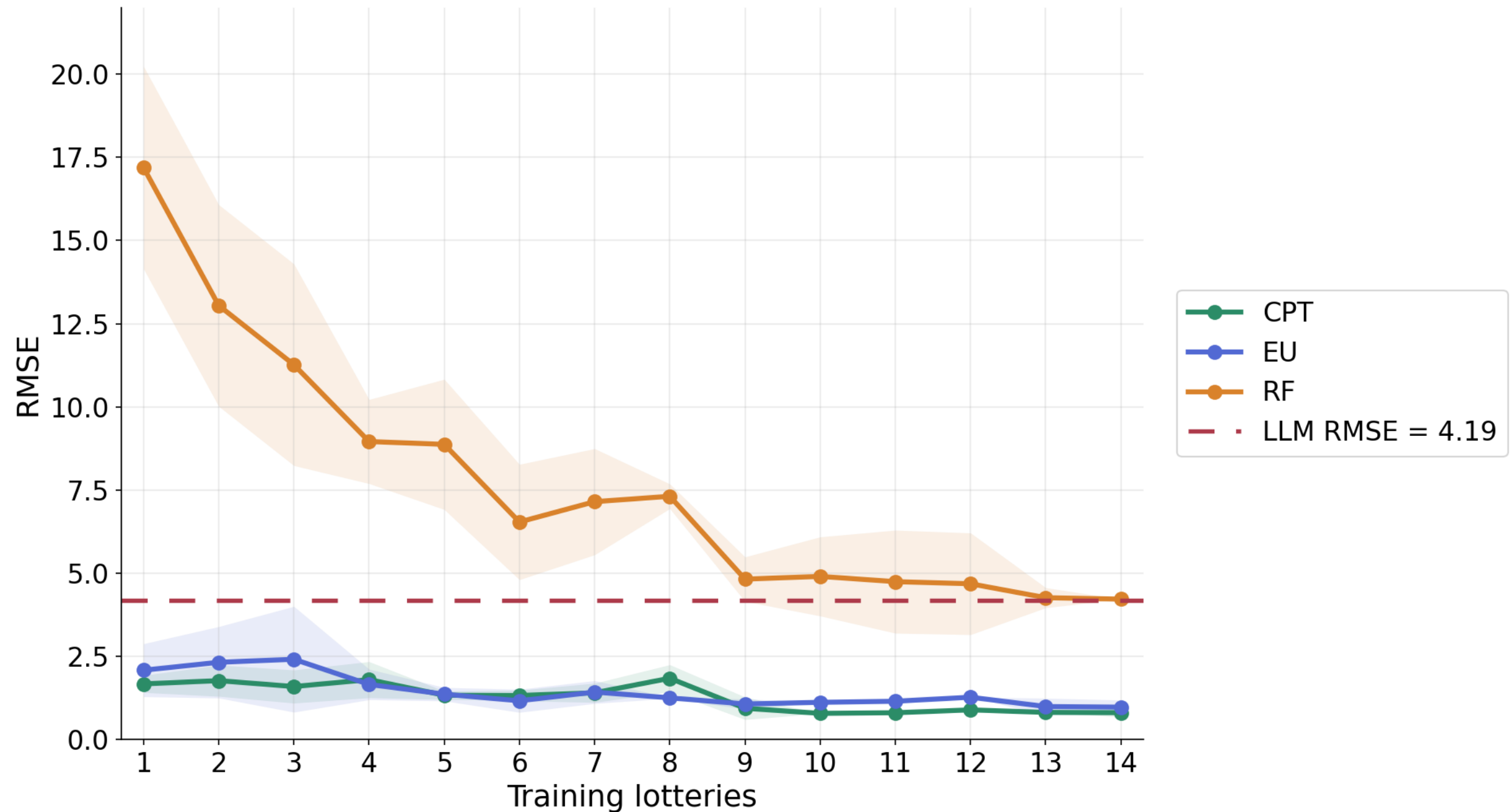
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N^* tells us: how many training lotteries do the domain-specific comparators need to beat the LLM's performance?

Example Country: Japan

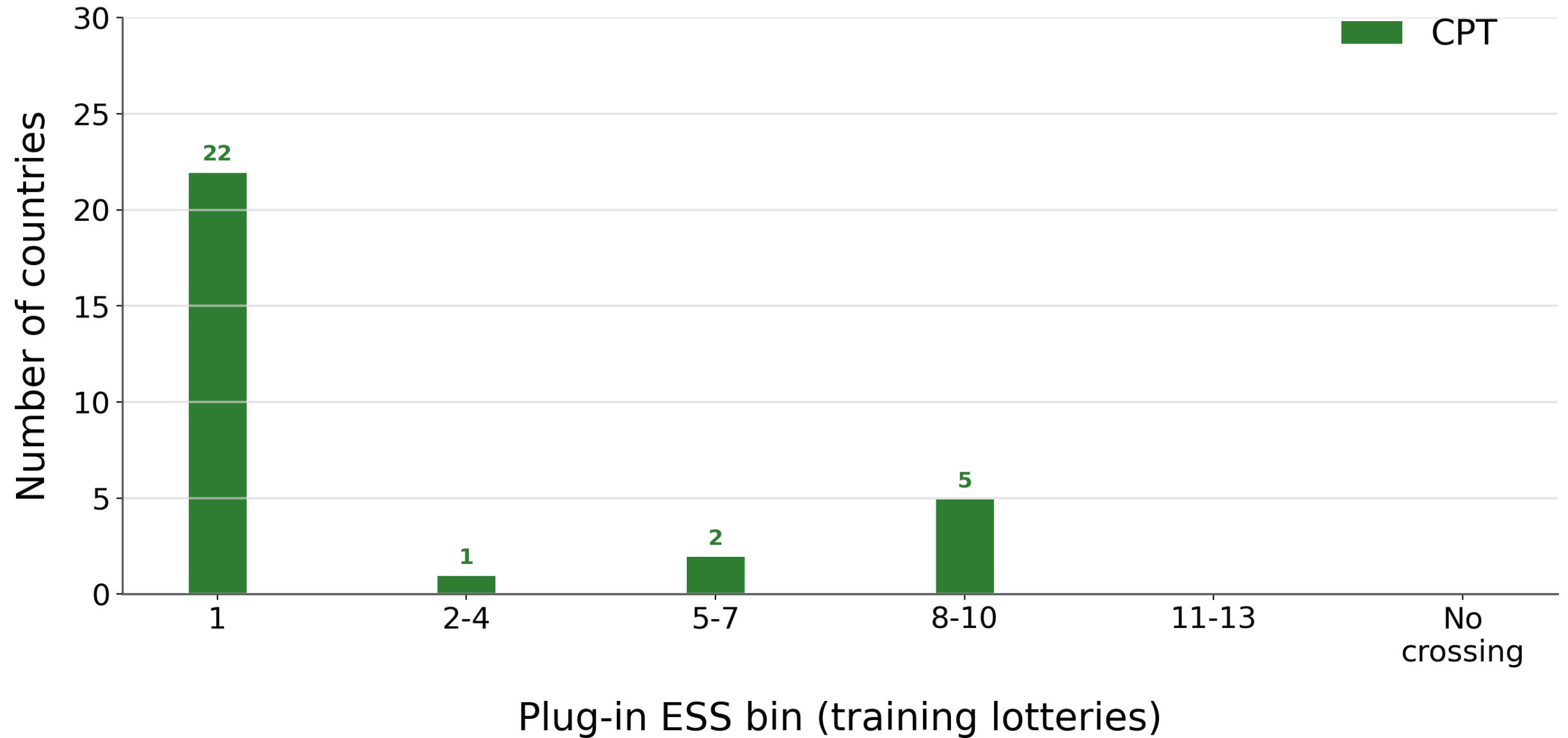


The Econ models beat GPT 5.4 when trained on a single lottery!

RF takes 14 lotteries to catch up

N^* Across Countries

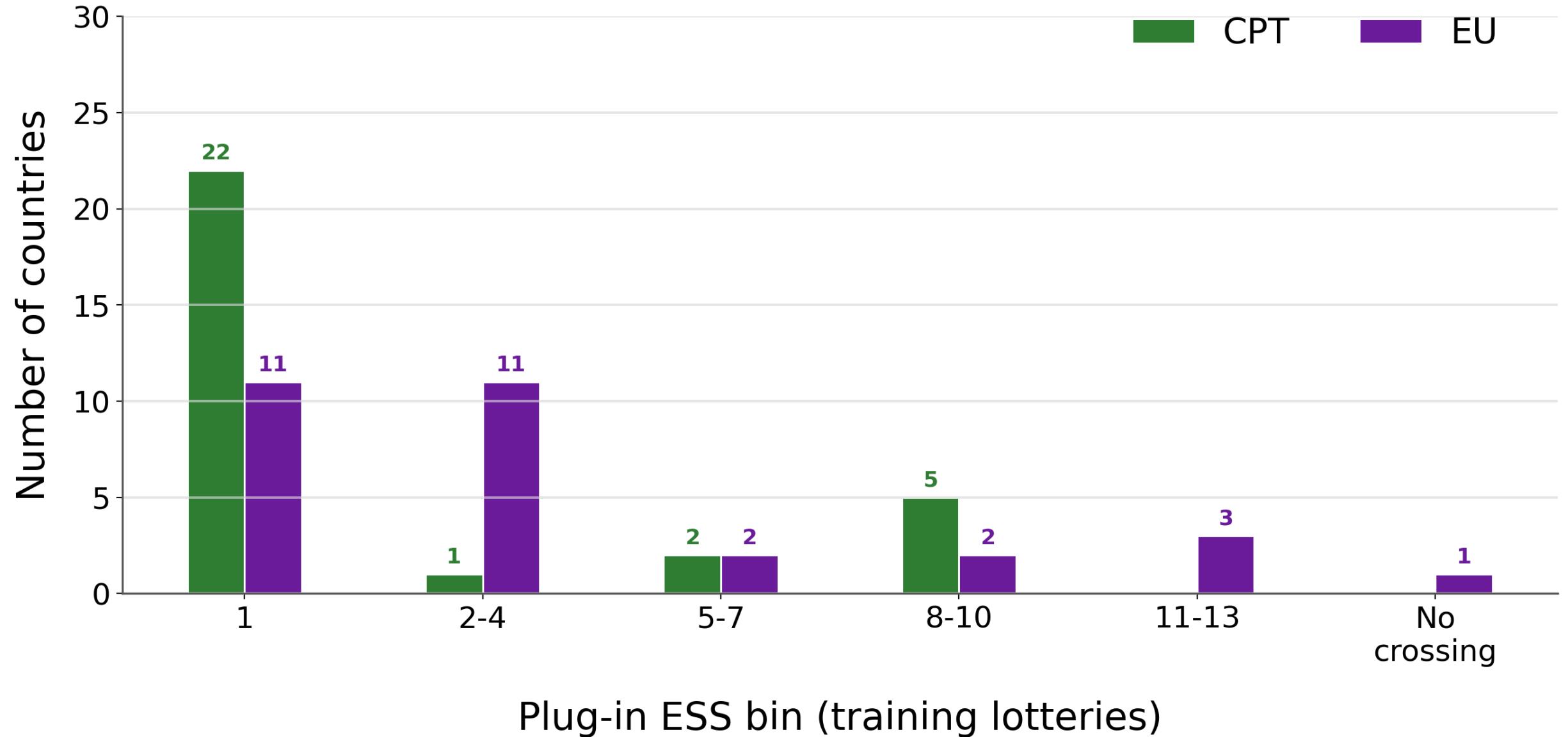
Median **CPT** ESS = 1



N^* Across Countries

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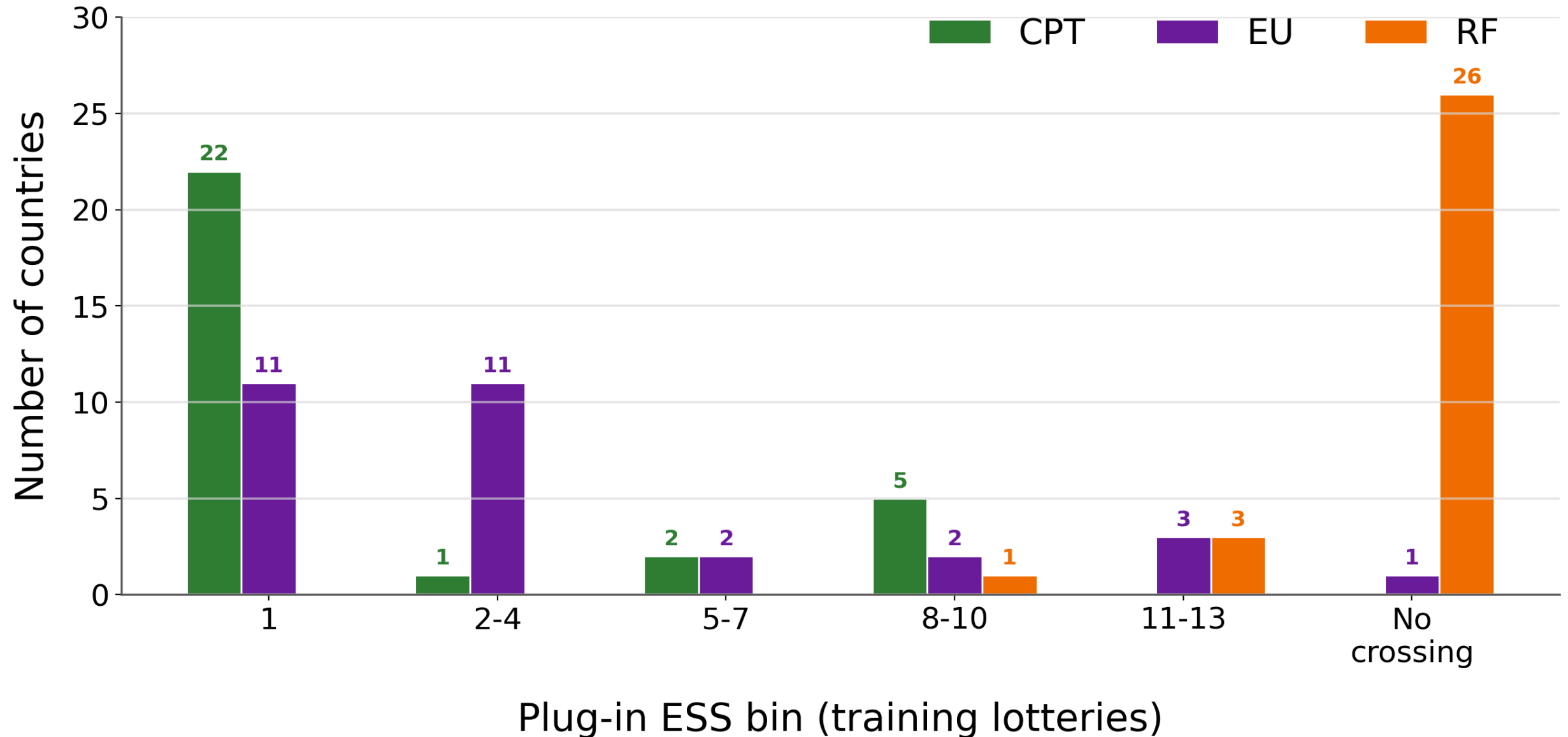
Median **EU** ESS = 4



N^* Across Countries

Median CPT ESS = 1

Median EU ESS = 4



Takeaways: (1) N^* is generally low for the economic models (but not for RF); (2) There is variation across countries

What Drives the Variation?

- The UN Human Development Index (HDI) is a composite of life expectancy, education, and per-capita income, normalized between 0 and 1
- LLM error tends to be lower in higher-HDI countries ($\rho = -0.44$), with a substantially weaker relationship for CPT ($\rho = -0.19$)
- N^* tends to be higher in higher-HDI countries ($\rho = 0.34$)
- Takeaway: LLMs are relatively more useful for prediction in more developed countries
 - Intuitive, but our measure helps us see this more clearly

Application 2:

Predicting Play in Games

Setting

Classification problem: given a 3x3 payoff matrix, predict the action most likely to be chosen by the row player.

	a_1	a_2	a_3		
a_1	25, 25	30, 40	100, 31	$\longrightarrow$	$\{a_1, a_2, a_3\}$ Y
a_2	40, 30	45, 45	65, 0		
a_3	31, 100	0, 65	40, 40		
X					

Our analysis: evaluate the ESS of GPT 5.4 relative to:

- A popular behavioral game theory model (Poisson Cognitive Hierarchy)
- Random Forest

Data

We consider two datasets:

- **Dataset A:** play in 86 games, aggregated across six experimental game theory papers (Wright and Leyton-Brown 13)

Data

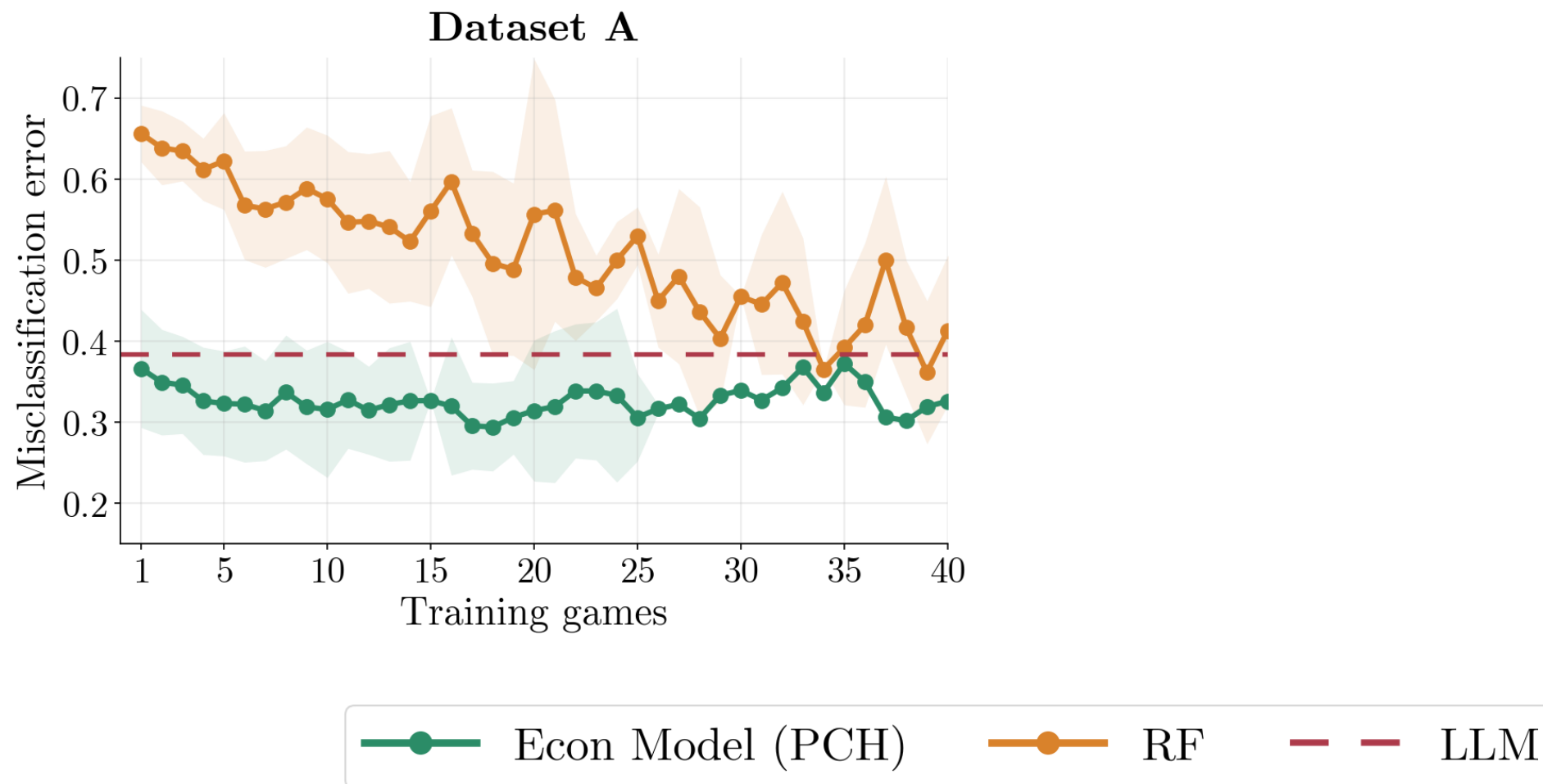
We consider two datasets:

- **Dataset A:** play in 86 games, aggregated across six experimental game theory papers (Wright and Leyton-Brown 13)
- **Dataset B:** play of 200 games, algorithmically generated to be strategically rich and different from those in Dataset A (Fudenberg and Liang 19)

Example:

	a_1	a_2	a_3	
a_1	90, 90	30, 80	45, 30	
a_2	80, 30	55, 55	37, 5	← Level 1
a_3	30, 45	5, 37	70, 70	

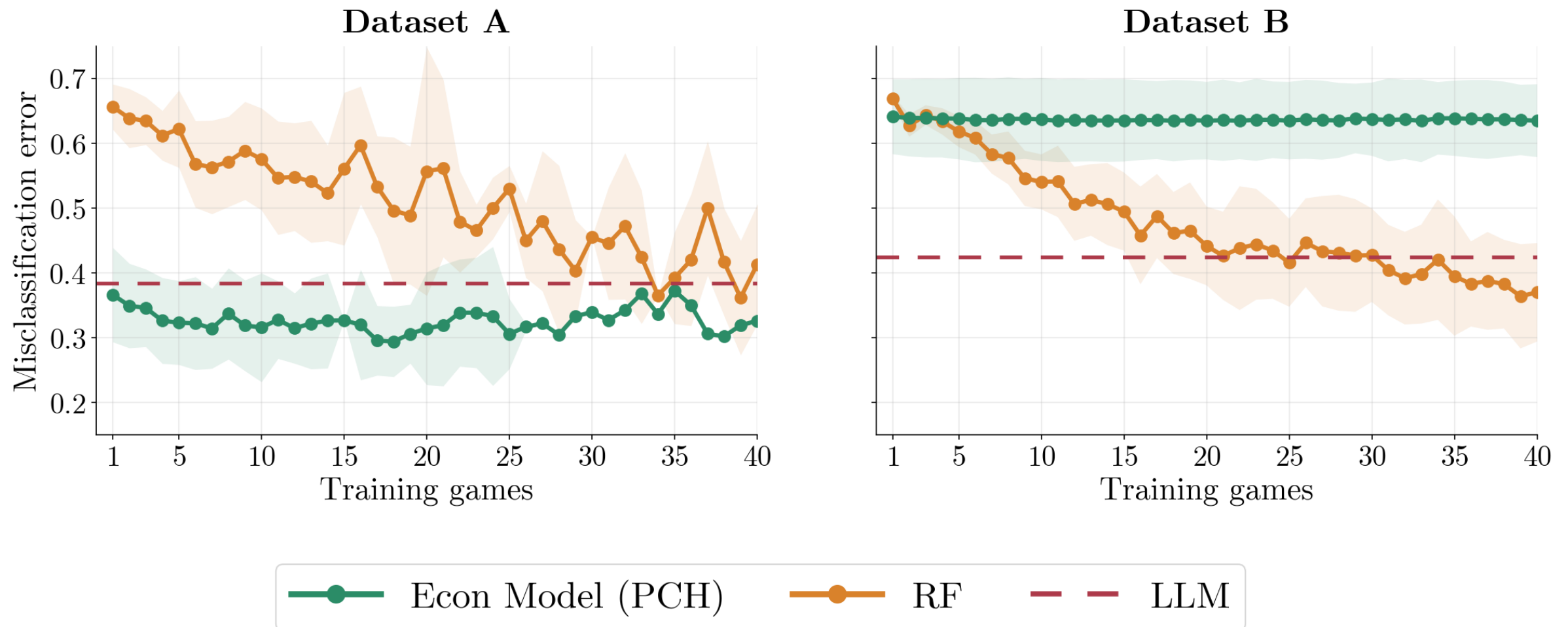
Results



On the lab games designed by experimental game theorists:

- ESS very low with respect to the economic model (Poisson Cognitive Hierarchy Model): plugin estimate $\hat{N}^* = 1$, confidence interval of $[1, \infty)$

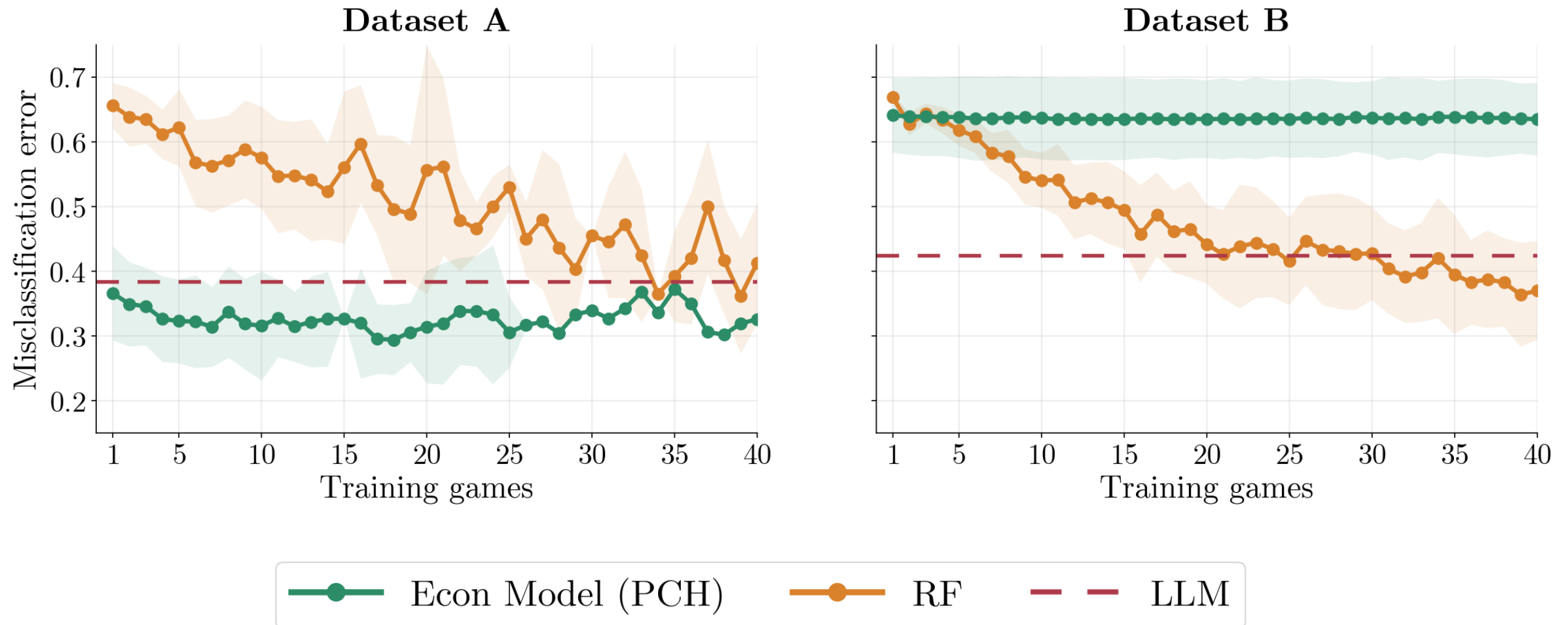
Results



In the new games:

- The LLM does nearly as well as it did on the lab games, but the economic model does much worse, and stays worse
- Eventually the ML algorithm surpasses the LLM (plugin estimate $\hat{N}^* = 25$, confidence interval of $[16, \infty)$)

Results



Takeaway: LLMs can be valuable when our theories fail

Application 3:

Survey Outcomes in PSID

Application 3: PSID

We predict outcomes in the Panel Study of Income Dynamics (PSID)

- Longitudinal survey that has followed US individuals and families since 1968, with data on economic, social, and health factors

Here is your persona: You are a 37-year-old male living in Tennessee in 2020. You work in architecture and engineering occupations in manufacturing industry. You have 4 years of work experience in the current job. You have completed 14 years of education. You are married and have 1 child. Your father has 12 years of education. Your mother has 12 years of education. Your race is white. Your health is excellent. You are Protestant.

Given your persona, do you own a house or apartment or do you rent? If you own, say 1; if you rent, say 5; if you neither own nor rent, say 8.

Application 3: PSID

We predict outcomes in the Panel Study of Income Dynamics (PSID)

- Longitudinal survey that has followed US individuals and families since 1968, with data on economic, social, and health factors
- Consider a PSID year beyond the LLM's knowledge cutoff
- Main takeaway: ESS (computed using RF) differs substantially across different outcome variables, e.g.,
 - **Smoking:** one-sided CI is $[1, \infty)$, plugin estimator $\hat{N}^* = 1$
 - **Hourly wage:** one-sided CI is $[17, \infty)$, plugin estimator $\hat{N}^* = 49$
 - **Home-ownership:** one-sided CI is $[590, \infty)$, no finite crossing

Conclusion

- Researchers/firms will increasingly use LLMs to predict economic behaviors and outcomes
- We need to understand when this is a good practice and when it is not
- We provide a measure and methodology to help with this evaluation
- The measure can help researchers understand and communicate the capability (or lack thereof) of LLMs across different domains

Conclusion

1. When is N^* high?

	LLM is good	LLM is bad
Econ model is good		
Econ model is bad/ doesn't exist		

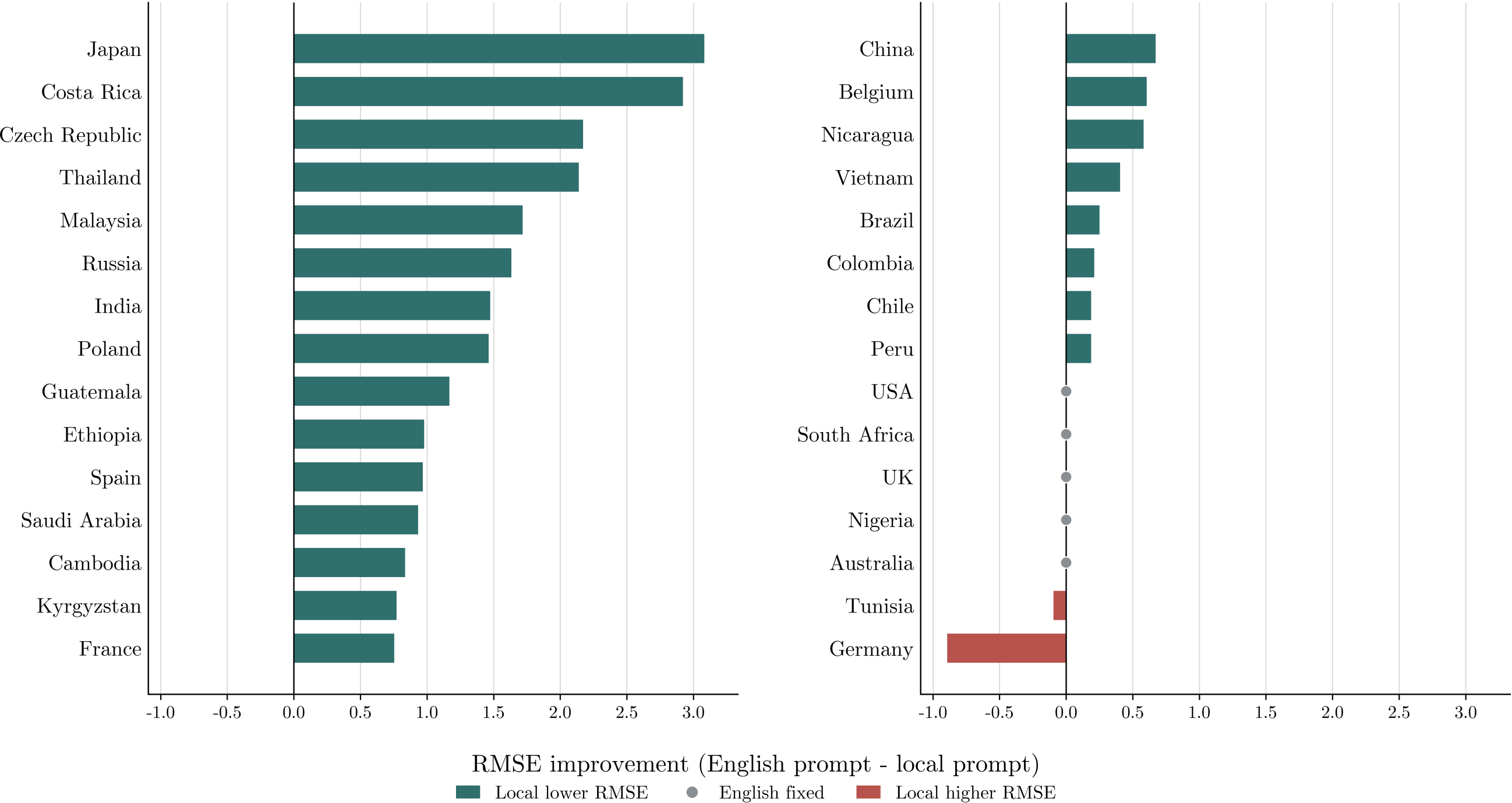
2. Where do LLMs have most value?

	High N^*	Low N^*
Data is costly		
Data is cheap		

Thank You

Change in LLM Error by Country

When is prompting in the local language most useful?



Variations on the LLM Prediction Rule

Alternative analyses:

- Use **DeepSeek** to make predictions

Method	GPT-5.4		DeepSeek	
	Median ESS	Crossings	Median ESS	Crossings
CPT	1	30/30	1	30/30
EU	4	29/30	2.5	30/30

LLM value **degrades**
(lower ESS)

Variations on the LLM Prediction Rule

Alternative analyses:

- Use **DeepSeek** to make predictions
- Ask the prompt **in Chinese**

Method	English prompt		Chinese prompt	
	Median ESS	Crossings	Median ESS	Crossings
CPT	1	30/30	6	27/30
EU	4	29/30	9	19/30

LLM value **improves**
(higher ESS)

Variations on the LLM Prediction Rule

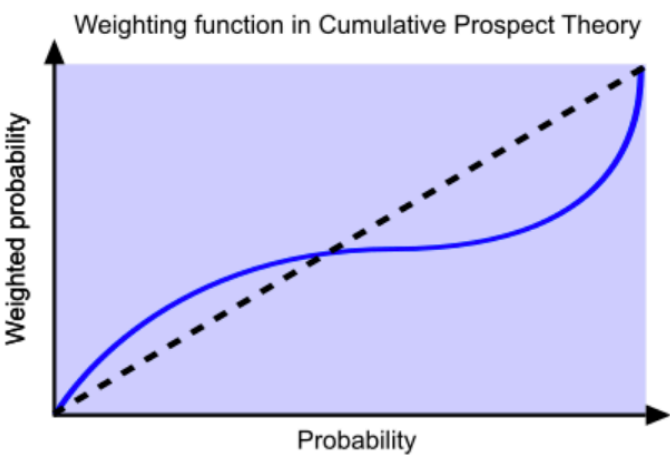
Alternative analyses:

- Use **DeepSeek** to make predictions
- Ask the prompt **in Chinese**
- Ask the prompt in **the primary language** of the country

Method	English prompt		Local prompt	
	Median ESS	Crossings	Median ESS	Crossings
CPT	1	30/30	5	27/30
EU	4	29/30	9.5	22/30

LLM value **improves**
(higher ESS)

Variations on the LLM Prediction Rule



Rough intuition:

- GPT 5.4 with English prompts is “too behavioral”

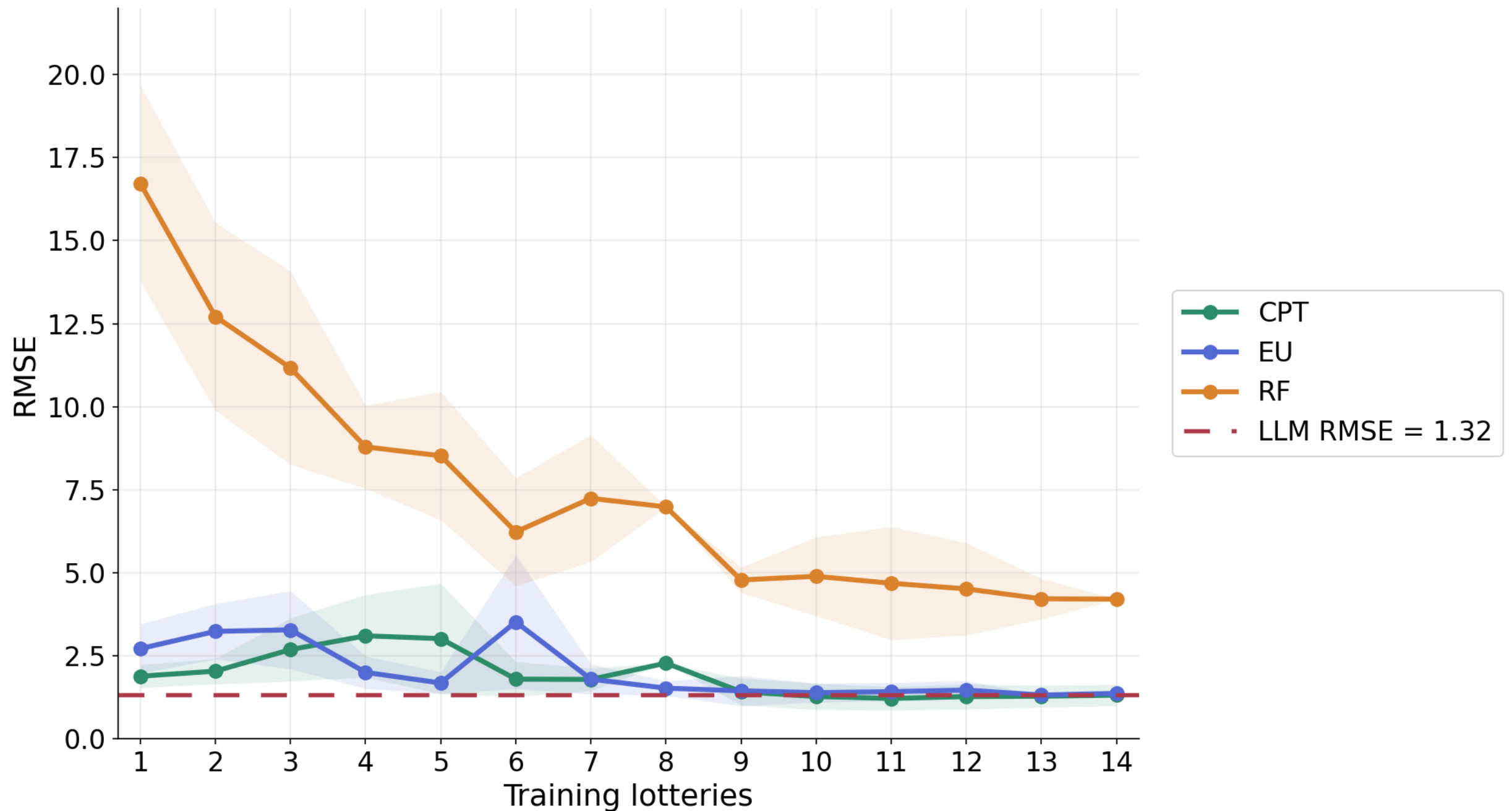
Rank	Model / prompt	Distance from expected value	Median fitted values	
			γ	δ
EU	Risk-neutral EU maximizer	0.000	1.000	1.000
1	GPT 5.4, Chinese prompt	1.119	0.865	0.889
2	GPT 5.4, Local language	1.393	0.838	0.875
3	GPT 5.4, English prompt	2.112	0.838	0.776
4	Deepseek, English prompt	2.931	0.600	0.713

probability distortion
parameters

Test for Each $H_{0,N}$

- $T_{N,n} \equiv \frac{\hat{e}_N^{CV} - \hat{e}_{LLM}}{\sqrt{(\hat{\sigma}_N^2 + \hat{\sigma}_{LLM}^2)/n}}, \hat{\sigma}_N^2 = N \cdot \widehat{Var}_{N,train} + \widehat{Var}_{N,test} + 2N \cdot \widehat{Cov}_N$
- $\widehat{Var}_{N,train} = Var_{D_N}(e(f_N))$ is the variance of the error of the prediction rule f_N (given randomness of the training sample D_N)
- $\widehat{Var}_{N,test} = Var_P(\mathbb{E}_{D_N}(\ell(f_N(X), Y)))$ is the variance of the expected loss of f_N on a new observation (X, Y) (given randomness of (X, Y))
- $\widehat{Cov}_N$ is the covariance of $e(f_N)$ and $e_N(X_1, Y_1)$ when (X_1, Y_1) is the first observation in the training sample D_N
- **Our test:** Reject $H_{0,N}$ if $T_{N,n} > c_\alpha$ for critical value c_α
- **Result:** This test is asymptotically valid (based on new asymptotic theory for cross-validated errors adapted for our fixed training size setting)

Highest ESS Country: Germany



The Econ models need more lotteries here to beat GPT 5.4

RF never catches up in the considered domain